

Advanced Topics in Computer Graphics I

Resampled Importance Sampling



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Problem: In 1 SPP rendering few paths actually find a light source. The image is mostly black with few bright spots.

Solution: Reuse information from previous frame and neighboring pixels to improve convergence. Achieved by reusing (good) light paths to increase the effective sample count.

Let $\hat{p} : \Omega \rightarrow \mathbb{R}$ be a potentially *unnormalized target distribution*.

Task: Importance sample $Y \sim \hat{p}$ from a sequence of M i.i.d. random variables $(X_i)_{i=1}^M \in \Omega$ with known PDF $p(X_i)$.

Picking $Y \in (X_i)_{i=1}^M$ leads to better importance sampler than any of the X_i 's.
Let s be a random variable of the chosen sample index, we set $Y = X_s$.

$$\Pr(s = i) = \frac{w_i}{\sum_{j=1}^M w_j},$$

with *resampling weights* $w_i = \frac{1}{M} \cdot \frac{\hat{p}(X_i)}{p(X_i)}$ for each X_i ,

$$\lim_{M \rightarrow \infty} p(Y) = \frac{\hat{p}(Y)}{\int \hat{p}(x) \, dx}$$

Note: $p(Y)$ is intractable, since we need to marginalize over all possible instances of X_i ! The resampled Y still leads to an unbiased estimator.



Basic RIS with i.i.d. samples can be performed in a streaming manner using $O(1)$ memory and $O(M)$ time using *reservoirs*.

Algorithm Streamed RIS with reservoir sampling

```
1: struct RESERVOIR
2:    $Y \leftarrow Y_\emptyset \in \Omega$                                 ▷ Initialize with empty/invalid sample.
3:    $M \leftarrow 0 \in \mathbb{N}$ 
4:    $\tilde{w}_{\text{sum}} \leftarrow 0 \in \mathbb{R}$                                 ▷ No  $\frac{1}{M}$  factor in  $\tilde{w}$ 's!
5:   function UPDATE( $X_i, \tilde{w}_i$ )                                ▷ Add sample to reservoir and increase size.
6:      $\tilde{w}_{\text{sum}} \leftarrow \tilde{w}_{\text{sum}} + \tilde{w}_i$ 
7:      $M \leftarrow M + 1$ 
8:     if rand() <  $\tilde{w}_i / \tilde{w}_{\text{sum}}$  then                                ▷ Generate random number in  $[0, 1]$ .
9:        $Y \leftarrow X_i$ 
10:    end if
11:  end function                                ▷  $Y$  is representative sample of all seen  $X_i$ 's.
12: end struct
```



Reservoirs can be combined in basic RIS without re-processing all samples. Let $r_1, \dots, r_k$ be reservoirs that have been constructed from samples $(X_i^{(1)})_{i=1}^{M_1}, \dots, (X_i^{(k)})_{i=1}^{M_k}$, respectively.

Algorithm Combining Reservoirs

```
1: function COMBINERESERVOIRS(Reservoirs  $r_1, \dots, r_k$ )  
2:    $s \leftarrow$  new Reservoir()  
3:   for all  $r \in \{r_1, \dots, r_k\}$  do  
4:      $s.\text{update}(r.Y, r.\tilde{w}_{\text{sum}})$   
5:   end for  
6:    $s.M \leftarrow \sum_{i=1}^k r_i.M$   
7:   return  $s$   
8: end function
```



Generalized Resampled Importance Sampling – Lin et al. (2022)





The goals of generalized RIS are to allow for:

- ▶ Potentially correlated random variables $(X_i)_{i=1}^M$.
- ▶ Potentially different domains, i.e. $X_i \in \Omega_i \neq \Omega$.
- ▶ Potentially resampled random variables $(X_i)_{i=1}^M$ with intractable PDFs $p_i(X_i)$.

Input samples might originate from neighboring pixels in time and space and might be the result of a previous resampling operation.

Given independent $(X_i)_{i=1}^M \in \Omega$ with *different* PDFs $p_i(X_i)$. i.e. $p_i(X_i) \neq p_j(X_i)$.

Introduce *resampling MIS weights* $m_i : \Omega \rightarrow \mathbb{R}$ with

$$m_i(x) \geq 0, \quad \sum_{i=1}^M m_i(x) = 1, \quad [p_i(x) = 0] \Rightarrow [m_i(x) = 0] \quad \forall x \in \Omega.$$

Resampling MIS weights $m_i(x)$ replace $\frac{1}{M}$ in *resampling weights* w_i :

$$w_i = m_i(X_i) \cdot \frac{\hat{p}(X_i)}{p_i(X_i)}$$

If $p_i(x) \neq 0$ for all i and $x \in \Omega$, then $m_i(x) = \frac{1}{M}$ leads to correct result. Otherwise balance-heuristic, power-heuristic or other MIS weights are required.

Note: This can typically no longer be streamed because the MIS weights $m_i(x)$ might depend on terms such as $p_i(X_j)$ for $i \neq j$.



Definition (Unbiased Contribution Weight)

A real-valued random variable $W \in \mathbb{R}$ is called an *unbiased contribution weight* for a random variable $X \in \Omega$ iff

$$\mathbb{E}[f(X) \cdot W] = \int_{\Omega} f(x) \, dx$$

for all integrable functions $f : \Omega \rightarrow \mathbb{R}$.

The expression $f(X) \cdot W$ generalizes $\frac{f(X)}{p(X)}$, i.e. $W = \frac{1}{p(X)}$ is valid for *known* $p(X)$.

Corollary

A real-valued random variable $W \in \mathbb{R}$ is an *unbiased contribution weight* for a random variable $X \in \Omega$ iff¹

$$\mathbb{E}[W \mid X] = \frac{1}{p(X)}$$

¹Section S.5.1 in https://graphics.cs.utah.edu/research/projects/gris/GRIS_supplemental.pdf

Given input samples $(X_i)_{i=1}^M \in \Omega$ and their unbiased contribution weights $W_i \in \mathbb{R}$, i.e. exact PDFs $p_i(X_i)$ are unknown.

The *resampling weights* w_i are re-defined as

$$w_i = m_i(X_i) \cdot \hat{p}(X_i) \cdot W_i$$

By construction, w_i is a random variable that unbiasedly estimates its previous definition.

$$\mathbb{E}[w_i] = m_i(X_i) \cdot \frac{\hat{p}(X_i)}{p_i(X_i)}$$

Question: What is unbiased contribution weight W_Y of the resampled Y ?

Let $g_i : \Omega \rightarrow \mathbb{R}$ be *unknown contribution functions* that are evaluated when input sample index $s = i$ is chosen, i.e. $Y = X_s$, where $\Pr[s = i] = w_i / \sum_{j=1}^M w_j$.

We want the *sum* of all sample-contributions to equal the target integral.

$$\mathbb{E}[f(Y) \cdot W_Y] = \int_{\Omega} f(y) \, dy \stackrel{!}{=} \sum_{i=1}^M \int_{\Omega} g_i(x) \, dx = \sum_{i=1}^M \mathbb{E}[g_i(X_i) \cdot W_i] = \mathbb{E}\left[\frac{g_i(X_i) \cdot W_i}{\Pr(s = i)}\right]$$

Example: Differently distributed $(X_i)_{i=1}^M$ with resampling weights $w_i = m_i(x) \cdot \hat{p}(x) \cdot W_i$. Setting $g_i(x) = m_i(x) \cdot f(x)$ yields

$$\begin{aligned} \mathbb{E}\left[\frac{g_i(X_i) \cdot W_i}{\Pr(s = i)}\right] &= \mathbb{E}\left[m_i(X_i) \cdot f(X_i) \cdot W_i \cdot \frac{\sum_{j=1}^M w_j}{w_i}\right] \\ &= \mathbb{E}\left[\frac{\cancel{m_i(X_i)} \cdot f(X_i) \cdot \cancel{W_i} \cdot \sum_{j=1}^M w_j}{\cancel{m_i(X_i)} \cdot \hat{p}(X_i) \cdot \cancel{W_i}}\right] \stackrel{!}{=} \mathbb{E}[f(Y) \cdot W_Y] \end{aligned}$$

Leading to the *unbiased contribution weight* for the resampled random variable Y :

$$W_Y = \frac{1}{\hat{p}(Y)} \sum_{j=1}^M w_j.$$

- ▶ Samples might be generated from neighboring pixels in space and time.
- ▶ Integration domain Ω_i of neighboring pixel i might differ from integration domain Ω of current pixel, for example due to different surface orientation.
- ▶ Relaxation of RIS to different integration domains Ω_i via shift mappings.

Definition (Shift Mapping)

A shift mapping $T_i : \Omega_i \rightarrow \Omega \cup \{Y_\emptyset\}$ is a function that is *bijective* on the domain $\mathcal{D}(T_i) \subseteq \Omega_i$ and the image $\mathcal{I}(T_i) \subseteq \Omega$ and the remaining domain $\Omega_i \setminus \mathcal{D}(T_i)$ is mapped to $Y_\emptyset$ an invalid sample $Y_\emptyset$.

Let $Y_i = T_i(X_i) \neq Y_\emptyset$, then $p(Y_i) = \left| \frac{\partial T_i(X_i)}{\partial X_i} \right|^{-1} p(X_i)$.

Remark: If $T_i(x) = Y_\emptyset$ we require $m_i(x) = 0$.



Let $X_i \in \Omega_i$ be an arbitrary input sample from a potentially different domain $\Omega_i \neq \Omega$ and $Y_i = T_i(X_i) \in \Omega$ be shifted sample in the current integration domain.

The *GRIS resampling weights* w_i are finally defined as

$$w_i := \begin{cases} m_i(Y_i) \cdot \hat{p}(Y_i) \cdot W_i \cdot \left| \frac{\partial Y_i}{\partial X_i} \right|, & \text{if } X_i \in \mathcal{D}(T_i) \\ 0, & \text{otherwise.} \end{cases}$$

This choice naturally “rejects” samples that fail to be shift-mapped, setting their resampling weight to zero.

The unbiased resampling weight W_i is defined wrt. integration of X_i over Ω_i and not over Ω . The expression $W_i \cdot \left| \frac{\partial Y_i}{\partial X_i} \right|$ is the unbiased contribution weight wrt. integration of Y_i over Ω .

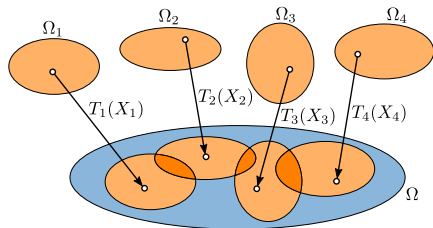
Using *contribution MIS weights* $c_i : \Omega \rightarrow \mathbb{R}$. With the GRIS resampling weights w_i we get the same unbiased contribution weights W_Y as before using contribution functions $g_i(X_i) = c_i(T_i(X_i)) \cdot f(T_i(X_i)) \cdot \left| \frac{\partial T_i(X_i)}{\partial X_i} \right|$,

$$W_Y = \left[\frac{c_i(Y)}{m_i(Y)} \right] \cdot \frac{1}{\hat{p}(Y)} \cdot \sum_{i=1}^M w_j$$

Remark: The MIS weight ratio is an additional source of variance. Choosing $c_i(x) = m_i(x)$ eliminates that and we get the same unbiased contribution weights as before

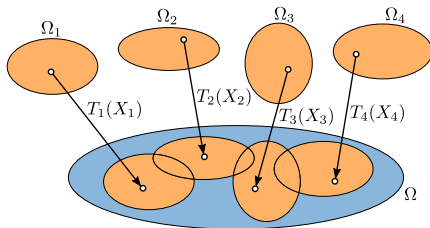
$$W_Y = \frac{1}{\hat{p}(Y)} \cdot \sum_{i=1}^M w_j.$$

Contribution MIS Weights Account for Overlaps



$$g_i(x) = \frac{1}{M} \cdot f(T_i(x)) \cdot \left| \frac{\partial T_i(x)}{\partial x} \right|$$

Combining shifted samples from different domains might lead to varying number of sampling strategies being able to yield samples in certain sub-domains.



$$g_i(x) = c_i(T_i(x)) \cdot f(T_i(x)) \cdot \left| \frac{\partial T_i(x)}{\partial x} \right|$$

Contribution MIS weights need to be chosen carefully to account for overlaps between different $T_i(\Omega_i)$!



Goal: Bounded contribution of final RIS sample Y , i.e. bounded resampling weights w_i .

Input samples $X_i \in \Omega_i$ are distributed according to a different unnormalized target distribution $\hat{p}_i : \Omega_i \rightarrow \mathbb{R}$.

Define “ $\hat{p}$ from i ” on the target domain as $\hat{p}_{\leftarrow i} : \Omega \rightarrow \mathbb{R}$ with

$$\hat{p}_{\leftarrow i}(y) = \begin{cases} \hat{p}_i(T_i^{-1}(y)) \left| \frac{\partial T_i^{-1}(y)}{\partial y} \right|, & \text{if } y \in \mathcal{I}(T_i) \\ 0, & \text{otherwise.} \end{cases}$$



Definition (Canonical Sample)

An input sample $X_i \in \Omega_i$ is *canonical* if $\Omega_i = \Omega$ and $p_i(x) \neq 0$ for all $x \in \Omega_i$, it uses the identity shift mapping $T_i(x) = x$, and is resampled to the unnormalized target distribution $\hat{p}_i = \hat{p}$.

- ▶ In GRIS *all* of Ω needs to be covered by an infinite stream of samples to guarantee asymptotic convergence.
- ▶ Provably covering Ω fully using samples from different domains Ω_i 's can be difficult.
- ▶ Add infinite stream of canonical samples to RIS to guarantee asymptotic convergence.
- ▶ “Fresh” canonical samples are supplemented by reused samples from other domains.



Theorem (Generalized Talbot MIS Weights)

Let $X_i \in \Omega_i$ with contribution weight W_i , $\hat{p}_i(X_i) \cdot W_i$ bounded and $Y_i = T_i(X_i) \in \Omega$ using the shift mapping T_i , then the resampling weights w_i are bounded using MIS weights $m_i : \Omega \rightarrow \mathbb{R}$

$$m_i(y) = \frac{\hat{p}_{\leftarrow i}(y)}{\sum_{j=1}^M \hat{p}_{\leftarrow j}(y)}.$$

Remarks:

- ▶ Analogous to balance heuristic.
- ▶ Computation cost in $O(M^2)$!



Let $\hat{p}_i(X_i) \cdot W_i \leq C_i$ for some constant $C_i \in \mathbb{R}_{\geq 0}$, and the resampling weight w_i

$$w_i = m_i(Y_i) \cdot \hat{p}(Y_i) \cdot W_i \cdot \left| \frac{\partial Y_i}{\partial X_i} \right| \leq C_i.$$

Multiplying with $\frac{C_i}{\hat{p}_i(X_i)W_i} \geq 1$ we have

$$w_i = m_i(Y_i) \cdot \frac{\cancel{\hat{p}(Y_i)} \cdot \cancel{W_i} \cdot C_i}{\hat{p}_i(X_i) \cdot \cancel{W_i}} \cdot \left| \frac{\partial Y_i}{\partial X_i} \right| \leq C_i,$$

which we also want to be bounded by the same C_i .

Find m_i such that the above inequality holds.

$$\Leftrightarrow m_i(Y_i) \leq \frac{\cancel{C_i}}{\cancel{C_i}} \cdot \frac{\hat{p}_i(X_i) \cdot \left| \frac{\partial Y_i}{\partial X_i} \right|}{\hat{p}(Y_i)} = \frac{\hat{p}_{\leftarrow i}(Y_i)}{\hat{p}_{\leftarrow k}(Y_i)},$$

where k is the index of an *arbitrary* canonical sample ($\hat{p}_k = \hat{p}_{\leftarrow k}$).

Summing over all used samples $1 \leq j \leq M$ in the denominator yields valid MIS weights

$$m_i(Y_i) = \frac{\hat{p}_{\leftarrow i}(Y_i)}{\sum_{j=1}^M \hat{p}_{\leftarrow j}(Y_i)} \leq \frac{\hat{p}_{\leftarrow i}(Y_i)}{\hat{p}_{\leftarrow k}(Y_i)}.$$

□



Theorem (Generalized Pairwise MIS Weights)

Let $X_i \in \Omega_i$ with contribution weight W_i , $\hat{p}_i(X_i) \cdot W_i$ bounded and $Y_i = T_i(X_i) \in \Omega$ using the shift mapping T_i , and R the set of canonical sample indices, i.e. if $i \in R$ then X_i is canonical. The resampling weights w_i are bounded using MIS weights $m_i : \Omega \rightarrow \mathbb{R}$

$$m_i(y) = \begin{cases} \frac{1}{M-|R|} \sum_{j \notin R} \frac{\hat{p}(y)}{|R| \cdot \hat{p}(y) + (M-|R|) \cdot \hat{p}_{\leftarrow j}(y)}, & \text{if } i \in R \\ \frac{\hat{p}_{\leftarrow i}(y)}{|R| \cdot \hat{p}(y) + (M-|R|) \cdot \hat{p}_{\leftarrow i}(y)}, & \text{if } i \notin R \end{cases}$$

given the above requirements.

Remarks:

- ▶ Usually $|R| \ll M$.
- ▶ Simple combination of $\hat{p}(Y_i)$ and $\hat{p}_{\leftarrow i}(Y_i)$ if $i \notin R$.
- ▶ Summation over $M - |R|$ terms if $i \in R$.
- ▶ Computation cost in $O(M \cdot |R|)$ instead of $O(M^2)$.



Recall the condition from Generalized Talbot MIS proof:

$$m_i(Y_i) \leq \frac{\hat{p}_{\leftarrow i}(Y_i)}{\hat{p}_{\leftarrow k}(Y_i)},$$

for an arbitrary canonical sample index $k \in R$, i.e. for which $\hat{p} = \hat{p}_{\leftarrow k}$.

For $i \in R$:

$$m_i(Y_i) \leq 1 = \frac{\hat{p}(Y_i)}{\hat{p}(Y_i)} = \frac{\hat{p}_{\leftarrow i}(Y_i)}{\hat{p}_{\leftarrow k}(Y_i)}$$

For $i \notin R$:

$$m_i(Y_i) = \frac{\hat{p}_{\leftarrow i}(Y_i)}{\hat{p}(Y_i) + (|R| - 1) \cdot \hat{p}(Y_i) + (M - |R|) \cdot \hat{p}_{\leftarrow i}(Y_i)} \leq \frac{\hat{p}_{\leftarrow i}(Y_i)}{\hat{p}_{\leftarrow k}(Y_i)},$$

since $1 \leq |R| \leq M$.

□

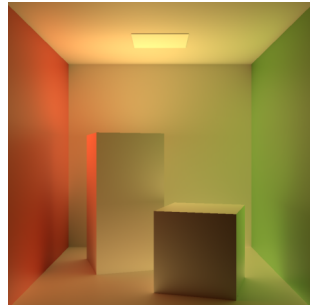
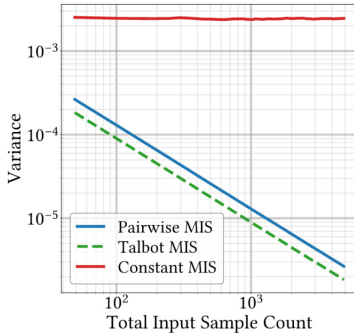


Figure: Single-sample integration error with GRIS, with increasing input samples from each pixel within a 7×7 reuse window. *Asymptotic zero-variance integration is realized with generalized Talbot and pairwise MIS weights, but not with constant MIS weights.* (Lin et al., 2022)

Problem: Streaming implementation is no longer possible with GRIS.

Want: Combine existing reservoirs without re-processing all samples.

The merge operation creates a new reservoir r_m from input reservoirs r_1 and r_2 . Y_{r_m} is resampled from Y_{r_1} and Y_{r_2} weighted with the respective M_{r_i} , and $M_{r_m} = M_{r_1} + M_{r_2}$. This is equivalent to resampling with *resampling MIS weights*

$$m_{r_i}(y) = \frac{M_{r_i}}{M_{r_1} + M_{r_2}}.$$

The interpretation of M as sample count is too strict. M_{r_i} is simply a relative weight between the two input reservoirs, which we call *confidence weight*.

Reservoir merging generalizes to GRIS with proper MIS weights by multiplying the $\hat{p}(y)$ terms with the corresponding M 's in the computation of resampling MIS weights $m_i(y)$.

Note: In practice, M_r is capped to a constant M_c , limiting the *confidence weight* of old samples, invalidating the old interpretation of M as sample count.



Importance of M -capping

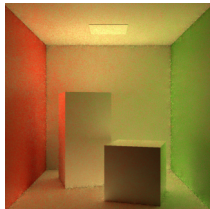


Without M -capping:

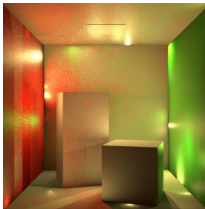
- ▶ Relative weight of new samples exponentially approaches zero.
- ▶ Convergence to wrong result due to correlations!

With M -capping:

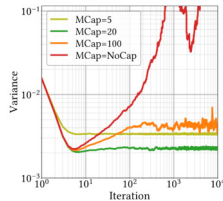
- ▶ Relative weight of new samples is at least $\frac{1}{M_c + 1}$.
- ▶ Result will not converge since $M \nrightarrow \infty$.
- ▶ Instead: Average of frames will converge in a still scene.



(a) M-cap: $M_c = 20$

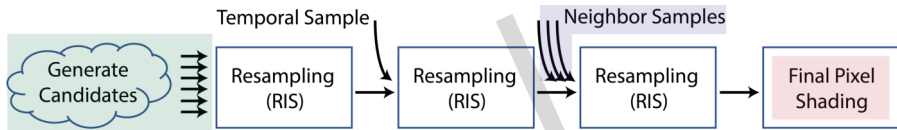


(b) No M-cap



(c) Effect of M-cap

Figure: Image after 10^4 iterations, with and without M-cap. (b) shows convergence to wrong result due to temporal correlations. (Lin et al., 2022)



(Wyman and Panteleev, 2021)

In each frame, run the following algorithm for each pixel:

Algorithm Resampling Pipeline

- 1: Generate independent $(X_{i,k}^t)_{k=1}^M \in \Omega$ with contribution weights $W_{X_{i,k}^t}$.
▷ Initial candidates
 - 2: $X_i^t = \text{resample}(X_{i,1}^t, \dots, X_{i,M}^t)$.
▷ Temporal reuse
 - 3: $Z_i = \text{resample}(X_i^t, Y_i^{t-1})$.
 - 4: Barrier: Synchronize execution for all pixels.
 - 5: Select (multiple) neighboring pixels $j_1, \dots, j_k$.
 - 6: $Y_i^t = \text{resample}(Z_i, Z_{j_1}, \dots, Z_{j_k})$.
▷ Spatial reuse
 - 7: Estimate $I_i^t \approx f_i(Y_i^t) \cdot W_{Y_i^t}$.
▷ Final shading
-

Runtime cost is $O(k + M)$, but each pixel sees $k \cdot M$ samples (+ temporal samples).



ReSTIR DI – Bitterli et al. (2020)



Simplified scenario: Only direct illumination (from many *diffuse* area light sources).

Resampling domain $\Omega \subseteq \mathbb{R}^3$ are simply positions on light source surfaces.

Valid light paths are $\bar{x} = (x_0, x_1, \mathbf{y})$, with camera vertex $x_0 \in \mathbb{R}^3$, shading vertex $x_1 \in \mathbb{R}^3$ and (resampled) light vertex $\mathbf{y} \in \Omega$.

The unnormalized target distribution $\hat{p}(\mathbf{y})$ is the unshadowed illumination

$$\hat{p}(\mathbf{y}) = \rho(x_0 \leftarrow x_1 \leftarrow \mathbf{y}) \cdot G(x_1 \leftrightarrow \mathbf{y}) \cdot L_e(\mathbf{y}).$$

Canonical samples are distributed proportional to emitted radiance $p_i(\mathbf{y}) \propto L_e(\mathbf{y})$.

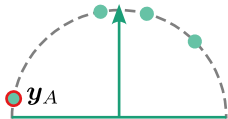
Identity shift mappings are used since $\Omega \supseteq \Omega_i$ for all samples including those from neighboring pixels q_i .

Target distribution of neighboring samples in spatio-temporal reuse is simply

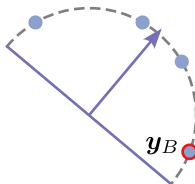
$$\hat{p}_{\leftarrow q_i}(\mathbf{y}) = \hat{p}_{q_i}(\mathbf{y}) = \rho(x_0^{(q_i)} \leftarrow x_1^{(q_i)} \leftarrow \mathbf{y}) \cdot G(x_1^{(q_i)} \leftrightarrow \mathbf{y}) \cdot L_e(\mathbf{y})$$

for a potentially different $x_0^{(q_i)}$ and $x_1^{(q_i)}$ at the source pixel q_i .

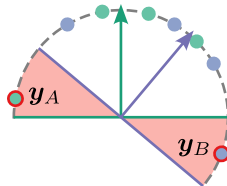
Consider two neighboring pixels A and B with different surface normal, with $\mathbf{y}_A$ and $\mathbf{y}_B$ resampled from 4 distant light samples each.



(a) Pixel A



(b) Pixel B



(c) Combined Pixels

If $\mathbf{y}_A$ from pixel A is reused in pixel B , $\hat{p}_B(\mathbf{y}_A) = 0$ and therefore the weight becomes zero and the sample is discarded successfully.

If $\mathbf{y}_B$ from pixel B is reused in pixel B , $\mathbf{y}_B$ could *not* be generated from pixel A ($\hat{p}_A(\mathbf{y}_B) = 0$), overestimating its sampling probability if uniform MIS weights $\frac{1}{M}$ are used naively!



ReSTIR DI uses uniform resampling MIS weights $m_i(\mathbf{y}) = \frac{1}{M}$.

Problem: Uniform resampling MIS weights $m_i(\mathbf{y}) = \frac{1}{M}$ violate the requirement that

$$[\hat{p}_{q_i}(\mathbf{y}) = 0] \Rightarrow [m_i(\mathbf{y}) = 0].$$

A naive implementation with uniform resampling MIS weights is *biased*!

Simple Solution (A): Use *different* uniform contribution MIS weights $c_i(\mathbf{y}) = \frac{1}{|Z|}$, where $Z = \{i \mid \hat{p}_{q_i}(\mathbf{y}) \neq 0\}$ is the set of sample indices which can produce $\mathbf{y}$ with non-zero probability.

Proposed Solution (B): Use contribution MIS weights $c_i(\mathbf{y}) = \frac{\hat{p}_{q_i}(\mathbf{y})}{\sum_{j=1}^M \hat{p}_{q_j}(\mathbf{y})}$,

i.e. generalized Talbot MIS weights.



Target distribution $\hat{p}$ does not include visibility information.

Problem: Even when (G)RIS converges ($M \rightarrow \infty$), there will be potentially severe noise due to visibility.

Solution: *Visibility reuse*. Visibility is usually locally coherent.

Perform visibility test *before spatial or temporal reuse* and discard occluded reservoir samples.

$\hat{p}_{q_i}(\mathbf{y})$ is used as proxy to estimate if $\mathbf{y}$ can be sampled from pixel q_i . If visibility reuse is enabled, we *also* need to evaluate the visibility for $\mathbf{y}$ from q_i because even if $\hat{p}_{q_i}(\mathbf{y}) \neq 0$ it will be discarded when occluded if visibility reuse is enabled, effectively setting $\hat{p}_{q_i}(\mathbf{y}) = 0$.

Biased version of algorithm: Don't check for visibility when evaluating contribution MIS weights!

Effect: Darkening when visibility is incoherent. Contribution MIS weights account for contribution from samples that will never exist.

- ▶ Bias only occurs if visibility is incoherent.
- ▶ Bias can be reduced by carefully selecting neighboring pixels for spatial (and temporal) reuse.
- ▶ Check for similar depth and normal when resampling.
- ▶ This avoids potentially bad samples causing bias with high probability.

Goal: Don't be unbiased, but minimize bias via heuristics for choosing neighbor pixels!

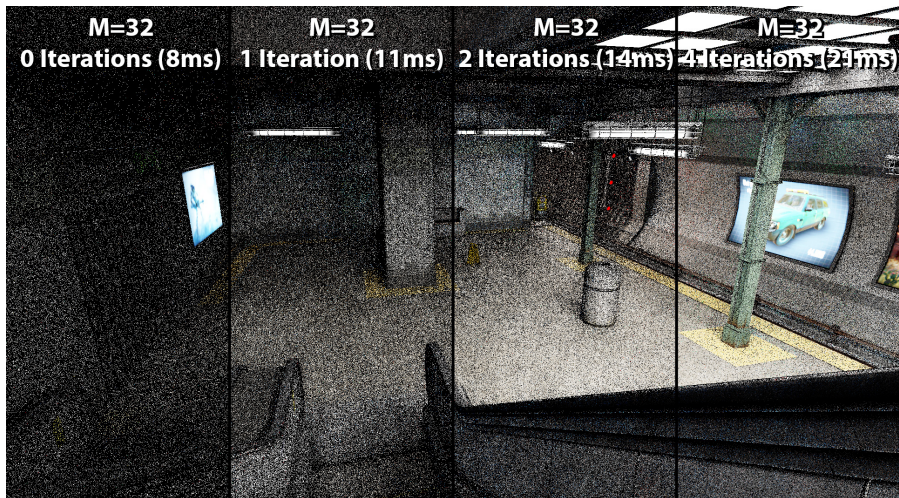


Figure: Starting from $M = 32$ initial candidates per pixel (left), we iteratively apply spatial reuse, gathering $k = 5$ neighbors at each step. (Bitterli et al., 2020)

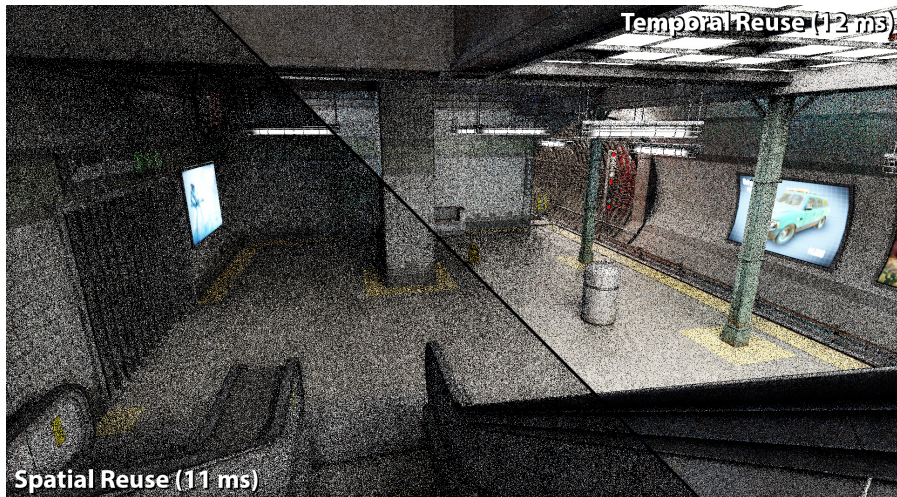


Figure: Compared to one iteration of spatial reuse alone (left, $M = 4$, $k = 5$), adding candidates from previous frames to candidates from the current frame can greatly increase the image quality (right, after 20 frames) with little added computational cost. (Bitterli et al., 2020)



ReSTIR GI – Ouyang et al. (2021)



Scenario: Consider $L_o(\mathbf{y})$ including reflections instead of $L_e(\mathbf{y})$.

Resampling domain $\Omega \subseteq \mathbb{R}^3$ includes any diffuse or light emitting surface.

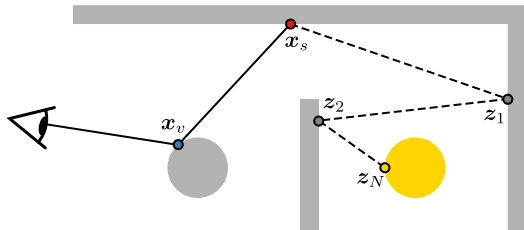
We call $\mathbf{x}_v = \mathbf{x}_1 \in \mathbb{R}^3$ *visible point* (from the camera), and $\mathbf{x}_s = \mathbf{y} \in \Omega$ *sample points* (representing a longer light path), replacing the light source samples from ReSTIR DI.

Canonical sample points $\mathbf{x}_s$ are sampled using BSDF importance sampling at $\mathbf{x}_v$. Outgoing radiance $L_o(\mathbf{x}_s)$ is estimated using 1 sample Monte-Carlo estimator.

Assumption: Perfectly *diffuse* BRDF at $\mathbf{x}_s$, i.e. $L_o(\mathbf{x}_s, \omega_o) = L_o(\mathbf{x}_s)$ for all ω_o .

Valid light paths are $\bar{\mathbf{x}} = (\mathbf{x}_0, \mathbf{x}_v, \mathbf{x}_s, \mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N)$, represented by $\mathbf{y} = \mathbf{x}_s \in \Omega$.

Rationale: If the scene is static, the contribution due to the subpath $\bar{\mathbf{z}} = (\mathbf{z}_1, \dots, \mathbf{z}_N)$ *after* the sample point $\mathbf{x}_s$ does *not* change in spatio-temporal resampling and can be ignored after initial sampling.



For the unnormalized target distribution $L_o(x_s)$ estimates the path throughput of $\bar{z}$.

$$\begin{aligned}\hat{p}(\mathbf{y}) &= \rho(\mathbf{x}_v, \omega_i, \omega_o) \cdot \cos(\theta_i) \cdot L_i(\mathbf{x}_v, \omega_i) \\ &= \rho(\mathbf{x}_0 \leftrightarrow \mathbf{x}_v \leftrightarrow \mathbf{x}_s) \cdot G(\mathbf{x}_v \leftrightarrow \mathbf{x}_s) \cdot L_o(\mathbf{x}_s)\end{aligned}$$

Better for spatiotemporal resampling (drop BRDF at $\mathbf{x}_v$):

$$\hat{p}(\mathbf{y}) = L_i(\mathbf{x}_v, \omega_i) = G(\mathbf{x}_v \leftrightarrow \mathbf{x}_s) \cdot L_o(\mathbf{x}_s)$$

- ▶ Resampling MIS weights $m_i(\mathbf{y})$ are used like in ReSTIR DI.
- ▶ Rejection heuristic for spatial reuse: Compare surface normal and depth of visible points $\mathbf{x}_v$.

Let $\mathbf{x}_v^q$ and $\mathbf{x}_v^r$ be the visible points of two different pixels q and r .

The sample point $\mathbf{x}_s^q$ for pixel q was created using BSDF importance sampling given $\mathbf{x}_v^q$ (as a direction wrt. solid-angle measure $d\omega$ and *not* position wrt. surface-area measure $d\mathbf{x}$).

When reusing a sample point $\mathbf{x}_s^q$ from a neighboring pixel q in r , we need to account for change of variables when mapping the corresponding incoming direction

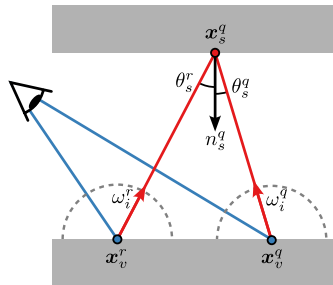
$\omega_i^q = (\mathbf{x}_s^q - \mathbf{x}_v^q) / \|\mathbf{x}_s^q - \mathbf{x}_v^q\|$ to $\omega_i^r = (\mathbf{x}_s^q - \mathbf{x}_v^r) / \|\mathbf{x}_s^q - \mathbf{x}_v^r\|$.

$$\hat{p}_{r \leftarrow q}(\omega_i^r) = \frac{\hat{p}_q(\omega_i^q)}{|J_{q \rightarrow r}(\omega_i^q)|}$$

$$|J_{q \rightarrow r}(\omega_i^q)| = \frac{\cos(\theta_s^r)}{\|\mathbf{x}_v^r - \mathbf{x}_s^q\|^2} \cdot \left[\frac{\cos(\theta_s^q)}{\|\mathbf{x}_v^q - \mathbf{x}_s^q\|^2} \right]^{-1}$$

with

$$\cos(\theta_s^q) = \langle \mathbf{n}_s^q | -\omega_i^q \rangle \quad \text{and} \quad \cos(\theta_s^r) = \langle \mathbf{n}_s^q | -\omega_i^r \rangle$$



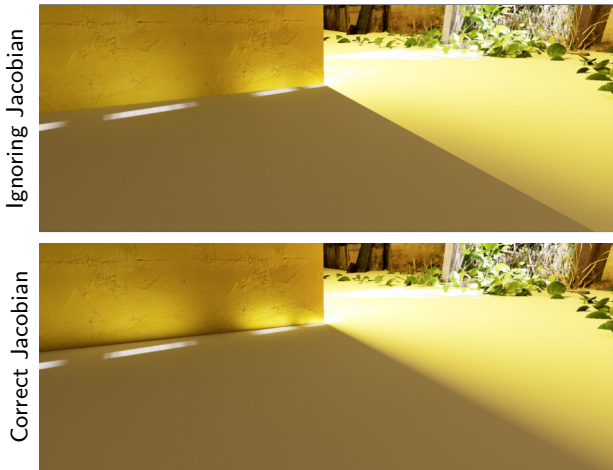


Figure: Effect of the Jacobian determinant. The wall receives sunlight and indirectly illuminates the floor. Top: Ignoring the Jacobian results in lighting discontinuities on the floor and overestimated lighting at the base of the wall. Bottom: Including the Jacobian corrects these artifacts. (Ouyang et al., 2021)



Note: Treating vertices x_s as samples and computing radiance in wrt. surface-area measure dx_s (including explicit geometry term) is equivalent to sampling corresponding *directions* ω_i (wrt. solid-angle measure $d\omega_i$) and then performing a change of variables from the hemisphere around x_v^q to x_v^r !

- ▶ View through **surface-area measure**:
Pixels q and r use same integration domain, but different target distributions.
- ▶ View through **solid-angle measure**:
Pixels q and r use different integration domains, i.e. local directions, requiring a change of variables, but the same target distribution.



Interjection: Solid-angle vs. Surface-area Measure



Using $G'_q := \frac{\cos(\theta_s^q)}{\|\mathbf{x}_s - \mathbf{x}_v^q\|^2} = G(\mathbf{x}_s \leftrightarrow \mathbf{x}_v^q) / \cos(\theta_i^q)$, i.e. geometry term without surface orientation at visible point.

Surface-area measure:

- Target distr.: (indep. of θ_i^q at $\mathbf{x}_v^q$)

$$\hat{p}_q(\mathbf{x}_s) = L_o(\mathbf{x}_s) \cdot G'_q$$

- Spatiotemporal resampling weight:

$$w_{r \leftarrow q}(\mathbf{x}_s) = \frac{\hat{p}_r(\mathbf{x}_s)}{\hat{p}_q(\mathbf{x}_s)} = \frac{G'_r}{G'_q}$$

Solid-angle measure:

- Target distr.: (indep. of θ_i^q at $\mathbf{x}_v^q$)

$$\hat{p}_q(\omega_q) = L_o(\mathbf{x}_s)$$

- Jacobian determinant:

$$|J_{q \rightarrow r}| = \left| \frac{\partial \omega_r}{\partial \omega_q} \right| = \left| \frac{G'_r}{G'_q} \right|$$

- Change of vars: $\hat{p}_{r \leftarrow q}(\omega_i^r) = \frac{\hat{p}_q(\omega_i^q)}{|J_{q \rightarrow r}|}$

- Spatiotemporal resampling weight:

$$w_{r \leftarrow q}(\omega_i^r) = \frac{\hat{p}_r(\omega_i^r)}{\hat{p}_{r \leftarrow q}(\omega_i^r)} = \frac{G'_r}{G'_q}$$



Problem: Sample points might have a long lifetime in reservoirs, which is problematic in dynamic scenes.

Solution: Periodically re-compute radiance $L_o(x_s)$ associated with the stored sample.

- ▶ Retrace light paths every few frames.
- ▶ **Important:** The light paths must be identical to the original path (if the scene did not change). Reuse the same random numbers during path generation.
- ▶ Include shadow ray from x_v to x_s in validation to detect moving occluders.
- ▶ If the radiance changed too much, discard the reservoir.
- ▶ Run sample validation instead of initial sample generation every $N = 6$ frames.



(a) Initial Lighting



(b) Without Sample Validation



(c) With Sample Validation



(d) Converged Result



Figure: Sample validation can effectively suppress bias caused by temporal reuse in dynamic scenes. Visible bias still remains after 12 frames (Bright spot on the left wall). With sample validation, the result rapidly adapts to changes in lighting. (Ouyang et al., 2021)



(a) Biased result



(b) 10 $\times$ difference

Figure: Bias caused by spatial reuse. Note that the bias is mainly in shadow areas due to visibility changes. The glossy floor worsens bias around the chair legs because the target function includes glossy reflection and tends to place more samples around the specular reflection direction. (Ouyang et al., 2021)

ReSTIR GI: Results

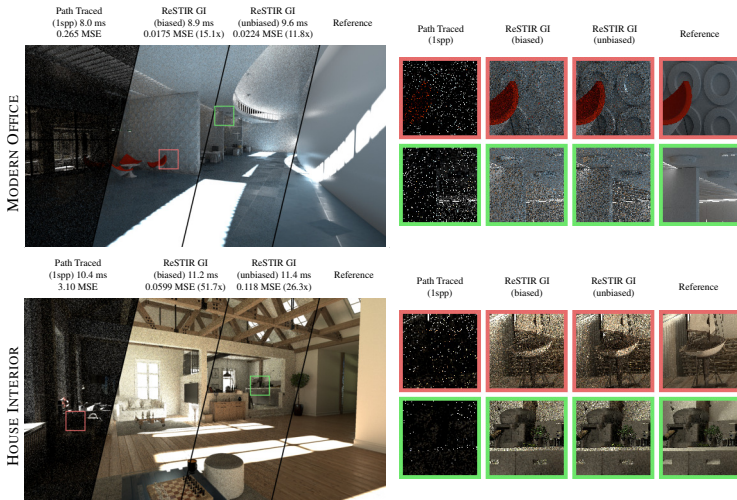


Figure: Comparison of roughly equal time renderings of standard path tracing and ReSTIR GI, with two-bounce indirect illumination. (Ouyang et al., 2021)



ReSTIR PT – Lin et al. (2022)





Generalization of ReSTIR to integration over full path-space, i.e. Path Tracing.

Integration domain $\Omega = \mathcal{P}$ are all light-carrying paths in the scene.

Samples $\bar{x} \in \Omega$ are arbitrary paths $\bar{x} = (x_0, \dots, x_n)$ of arbitrary length n connecting light sources x_n to the sensor x_0 .

For pixel q_i we have a filter kernel $h_i(\bar{x})$ defining the contribution of path $\bar{x}$ to pixel q_i . Using a simple box filter, integration is restricted to paths that exactly pass through pixel q_i :

$$I_i = \int_{\Omega} h_i(\bar{x}) f(\bar{x}) \, d\bar{x} = \int_{\Omega_i} f(\bar{x}) \, d\bar{x}.$$

The domain $\Omega_i \subseteq \Omega$ is the set of all paths passing through pixel q_i .

Note: Since $\Omega_i \neq \Omega_j$, reuse of a path $\bar{x} \in \Omega_i$ from pixel q_i in pixel q_j requires path modifications. (i.e. shift mappings)



The unnormalized target distribution $\hat{p}_{q_i}(\bar{\mathbf{x}})$ in pixel q_i is

$$\hat{p}_{q_i}(\bar{\mathbf{x}}) = |f(\bar{\mathbf{x}})|$$

or a cheap approximation thereof. . .

Note: Visibility is always taken into account here.

Canonical samples are generated via regular path tracing with next-event estimation.

- ▶ A single path-tracing sample creates multiple light-sensor paths (1-2 of each path length, via BSDF sampling and next-event estimation, respectively).
- ▶ Contributions are *not* directly accumulated like in regular path tracing.
- ▶ Paths are resampled using a reservoir to produce a single (canonical) candidate.



Problem: Temporal correlations slow convergence in progressive renderer.

Solution: Render independent frames with spatial-only reuse.

Two-pass algorithm:

1. Compute N independent “sub-frames” of cross-pixel reuse.
2. Average resulting N images of each “sub-frame”.

Each individual sub-frame might converge to wrong result, but on average the result is unbiased. Thus averaging multiple independent sub-frames still converges to the correct result.

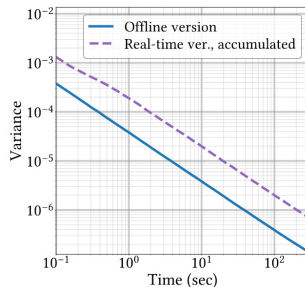


Figure: The offline method (blue) turns off temporal reuse, which converges faster when averaging frames; it avoids the frame-to-frame correlation introduced by temporal reuse. (Lin et al., 2022)



The original path $\bar{x}$ in pixel q_j is called *base path*, the shifted path $\bar{y} = T(\bar{x})$ in pixel q_i is called *offset path*.

Let x_0 be the sensor position, x_1 is then defined by tracing through pixel q_j .

Need heuristic for vertices y_k of shifted path for all $k \geq 2$.

Desired properties of shift mapping T are:

$$f(T(\bar{x})) \approx f(\bar{x}) \qquad \left| \frac{\partial T(\bar{x})}{\partial \bar{x}} \right| \approx 1$$

Offset path is constructed vertex-by-vertex using local decisions.

Decide y_{i+1} given y_{i-1}, y_i and x_{i-1}, x_i, x_{i+1} .



Vertex copy (Reconnection):

If x_i, x_{i+1} and y_i are rough materials, choose $y_{i+1} = x_{i+1}$.

Half-vector copy:

On specular materials copying the half-vector can be a good choice. Copy half-vector in local tangent space and retrace y_{i+1} as the reflection or refraction direction.

Direction copy:

If $\|x_{i+1} - x_i\|$ is large, the outgoing direction at x_i can simply be copied to y_i in world space to retrace y_{i+1} .

Random replay:

Copy the base path's random numbers to retrace y_{i+1} . Similar to half-vector copy on specular materials, direction copy for environment maps or reconnection for light sources.

Manifold exploration: (Jakob and Marschner, 2012)

If x_{i+1} is not reconnectable due to specularities, find next connectable vertex x_{i+k} and apply manifold exploration on $(y_i, x_{i+1}, \dots, x_{i+k})$.

Problem: Sequential construction of offset path might fail.

- ▶ E.g. half-vector copy might map refraction onto total-internal reflection; but (total-internal) reflections are always mapped to reflections.

Solution: Restrict shift-mapping domain. Not all paths can be shifted.

In practice we have symmetric shift mappings:

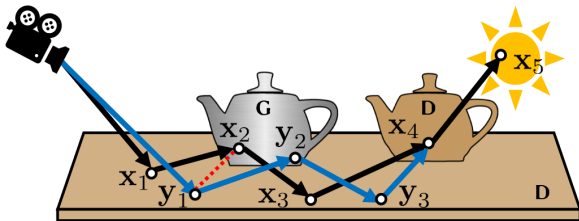
$$T_{i \leftarrow j}(\bar{x}) = \bar{y} \Leftrightarrow T_{j \leftarrow i}(\bar{y}) = \bar{x}$$

Remark: Restricting domain causes noise, because samples are “wasted”, but not guaranteeing bijectivity introduces significant bias.

Reconnection Shift: Always reconnect at first indirect vertex, i.e. $y_2 = x_2$. Implicitly done in ReSTIR GI (Ouyang et al., 2021).

Hybrid random replay and reconnection: If a vertex is *not* classified as reconnectable, use random replay. Otherwise reconnect at the *first* reconnectable vertex.

- ▶ Naive implementation: Need to store all potentially reconnectable base-path vertices for arbitrary offset path.
- ▶ Optimization: Base path determines single candidate for reconnection, where x_i and x_{i+1} are both rough materials. Reconnection must happen here or not at all.
- ▶ Rationale: If base and offset paths are reasonably similar, they should agree on reconnection index. This is also forced by bijectivity requirement!





ReSTIR PT: Reconnection vs Hybrid Shift Mapping

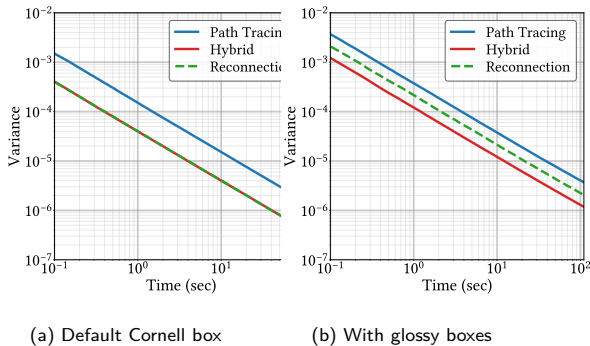


Figure: Error comparison of ReSTIR PT with the reconnection shift (blue) and the hybrid shift (green). Both shift mappings are good for rough scenes (left), but the hybrid shift (green) is more suited for scenes with glossy surfaces (right). (Lin et al., 2022)



In ReSTIR PT a reservoir stores:

- ▶ Unbiased contribution weight W_r
- ▶ Confidence weight M_r

Instead of a candidate path $\bar{y}_r$, for shift mapping we need:

- ▶ Reconnection vertex x_{i+1}
- ▶ Seed for random replay

Distance condition: Geometry term is unbounded for short path segments. Increased noise when reconnecting near-by vertices.

Only reconnect if the participating path segment lengths are above a threshold

$$\|x_{i+1} - x_i\| \geq d_{\min} \quad \text{and} \quad \|x_{i+1} - y_i\| \geq d_{\min}.$$

Lobe-specific condition: Only evaluate single BSDF lobe (e.g. only diffuse or specular) at each path vertex.

Classify vertex x_i as rough if ...

- ▶ The roughness of the selected lobe is above a given threshold.
- ▶ Next-event estimation is performed at x_i and the roughness of at least one lobe is sufficiently rough.
- ▶ All light vertices are treated as rough.
- ▶ The sensor vertex is treated as specular.

Combine each path $\bar{x} = (x_0, \dots, x_d)$ with sequence of lobe indices $\bar{\ell} = (\ell_1, \dots, \ell_{d-1})$ of interior vertices.

Each ℓ_i represents the index of the selected lobe at vertex x_i , or special symbol $\ell_{d-1} = \mathcal{N}$ if x_d was reached via next-event estimation at x_{d-1} .

Partial path-contribution function $f_{\ell}(\bar{x})$ evaluates lobe ℓ_i at vertex x_i or all lobes if $\ell_i = \mathcal{N}$.

$$f(\bar{x}) = \sum_{\bar{\ell}} \lambda_{\chi(\ell_{d-1}=\mathcal{N})}(\bar{x}) \cdot f_{\ell}(\bar{x})$$

- ▶ $\chi(\ell_{d-1} = \mathcal{N}) \in \{0, 1\}$ indicates if BSDF sampling or next-event estimation was performed to reach the last vertex.
- ▶ $\lambda_{\{0,1\}}(\bar{x})$ are the MIS weights of either strategy (balance or power heuristic) for the given path.

Advantage: Extended path-space *enables* lobe-specific operations/conditions.

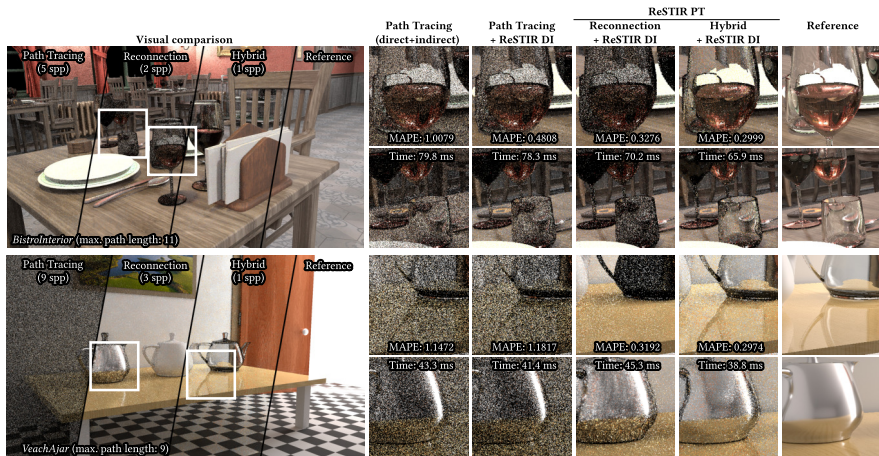


Figure: Comparing path tracing and two variants of ReSTIR PT (reconnection and hybrid shift) for real-time rendering. (Lin et al., 2022)



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